

BIOL-1 Practice Test 04

Exam 03 Preparation (Modules 12–16)

Instructions: This practice test covers material from **Module 12** (Darwin and Evolution) through **Module 16** (Capstone Systems Synthesis). Answer all questions to the best of your ability to prepare for Exam 03.

Part A: Multiple Choice (25 questions)

Choose the best answer for each question.

Module 12: Darwin and Evolution

1. In the logic of natural selection, which observation states that more offspring are produced than can usually survive to reproduce?
 - A) Differential reproductive success
 - B) Overproduction of offspring
 - C) Inheritance of acquired characters
 - D) Stabilizing selection only
2. A trait increases **evolutionary fitness** in a given environment if individuals with that trait tend to:
 - A) Look larger than their competitors
 - B) Contribute more alleles to the next generation than competitors do
 - C) Acclimate in one lifetime to match the parent phenotype
 - D) Survive long enough to be counted, even with zero offspring
3. Which statement is **most** consistent with **descent with modification**?
 - A) Each species was created in its final form and does not change
 - B) Lineages change over time; related species share ancestors with modification
 - C) Only fossils evolve, not living species

D) The strongest individual in a species can change the population's heritable gene pool in one lifetime, without reproduction

4. Homologous forelimb bones in a human, a cat, and a whale are used as evidence for evolution because they:

- A) Show identical functions in all three species
- B) Suggest a shared ancestry with structural change for different uses
- C) Prove that whales evolved from modern humans
- D) Disprove the existence of a fossil record

5. A population of moths in a clean forest is mostly light-colored; in a sooty industrial region the population becomes mostly dark. The clearest **selection** label for that shift, if the environment favors the dark extreme, is:

- A) Stabilizing selection
- B) Disruptive selection
- C) Directional selection
- D) The founder effect

6. Which option best states why **populations** evolve, not **individuals**?

- A) Individuals can change their DNA by lifting weights
- B) Evolution is a change in allele **frequencies in a population** across generations
- C) One hungry bird cannot affect allele frequencies at all (bird effects are not evolution)
- D) Natural selection has nothing to do with heritable variation

Module 13: How Populations Evolve

7. The definition of **microevolution** used in this course is:

- A) A change in an individual's size during one lifetime
- B) A **change in allele frequencies** in a population over time
- C) The origin of a new phylum in one generation
- D) A synonym for any mutation in a body cell only

8. A few finches from the mainland start a new island population. The island gene pool reflects only the alleles the founders brought. This is:

- A) The bottleneck effect
- B) The **founder** effect (genetic drift)

- C) Stabilizing selection on beak size
- D) Convergent evolution

9. A volcanic eruption kills 95% of a squirrel population at random, then survivors repopulate.

The loss of many alleles from the pre-eruption pool is:

- A) A founder effect in a new island colonization from another continent only
- B) A **bottleneck** (genetic drift)
- C) A proof that survivors were the “most fit” for every possible future environment
- D) Interspecific competition in the absence of intraspecific variation

10. In **stabilizing** selection on a trait (for example, birth weight in some human data sets):

- A) Both extreme phenotypes are favored
- B) The **intermediate** phenotype is favored; extremes are selected against
- C) The population’s average always moves to one tail of the range
- D) Allele frequencies must stay exactly at Hardy–Weinberg equilibrium forever

11. The Hardy–Weinberg equilibrium model is useful mainly because it:

- A) Describes a real population that never evolves
- B) Serves as a **mathematical baseline** to measure evolutionary change in real data
- C) Replaces the biological species concept
- D) Guarantees $p^2 + 2pq + q^2 = 1$ in every generation without assumptions

12. The movement of alleles into or out of a population through **migration and mating** is:

- A) Non-random mating only
- B) **Gene flow**
- C) Genetic drift in an infinitely large population
- D) A postzygotic barrier to reproduction

Module 14: Macroevolution

13. Under the **biological species concept**, two groups are different species if they:

- A) Look different in a field guide, regardless of interbreeding
- B) Are **reproductively isolated** in nature (they do not interbreed to produce viable, fertile offspring **between** the two groups, while within each group interbreeding can occur)
- C) Have different number of cells
- D) Live on different trees in the same forest, without any other test

14. Fireflies of two species use different **flash** patterns, so they rarely attempt mating. This isolating mechanism is best classified as:

- A) Postzygotic reduced hybrid fertility
- B) A **prezygotic behavioral** barrier
- C) A mechanical postzygotic barrier
- D) The bottleneck effect

15. A mule, the hybrid of a horse and a donkey, is typically sterile. This is an example of:

- A) A prezygotic gametic barrier
- B) A **postzygotic** barrier to gene flow (reduced hybrid fertility)
- C) Stabilizing selection in horses
- D) The founder effect

16. A new mountain range **physically splits** a population; over long time, the two lineages no longer interbreed. The classic geographic mode of speciation is:

- A) Sympatric speciation in one meadow
- B) **Allopatric** speciation
- C) The bottleneck effect in both lineages on purpose
- D) A requirement that drift never occurs

17. A plant can form a new, reproductively isolated **polyploid** lineage in a few generations, often in the same area as a parent type. This route is a frequent example of:

- A) The bottleneck effect only
- B) **Sympatric** speciation (e.g. via polyploidy)
- C) A prezygotic postzygotic hybrid
- D) Only macroevolution in animals, not plants

18. The statement “macroevolution is different chemistry from microevolution” is **false** in this course because:

- A) No DNA is involved in speciation
- B) The same **population**-level processes (selection, drift, flow, etc.) can accumulate to reproductive isolation and larger-scale diversity
- C) All macroevolution happens in a single meiosis
- D) The fossil record is irrelevant to modern biology

Module 15: Population and Systems Ecology

19. A population introduced to a new habitat with abundant resources for a time shows growth that **accelerates** (as long as the ceiling is not yet reached). That early phase is classic for:

- A) A flat line at carrying capacity from year one only
- B) An **exponential (J-shaped)** phase of growth
- C) A carrying capacity of zero by definition
- D) Only density-independent regulation, never K

20. A population levels off as resources become scarce, approaching a relatively stable size. That ceiling is:

- A) A genetic bottleneck
- B) **Carrying capacity (K)** in a logistic (S) model
- C) A proof that 100% of energy moves to the next level
- D) A synonym for the founder effect

21. A fungal disease that spreads more easily in **dense** host stands is an example of a factor often:

- A) **Density-dependent** in its impact on the population
- B) Only density-independent like wind
- C) Unrelated to per-capita rates
- D) Never biological

22. A severe freeze kills a **similar fraction** of a plant population across sparse and dense patches (the freeze does not “care” about crowding the way a contagion might). Treated as a **limiting** event in textbook classification, the freeze is **most** like:

- A) Only density-dependent competition for light
- B) A **density-independent** disturbance (still harmful to the population)
- C) A trophic level
- D) A biogeochemical cycle for nitrogen only

23. The **~10% rule** between trophic levels implies that, compared to eating **producers** directly, a diet of **higher** trophic levels (many steps) to deliver the same human calories will tend to need:

- A) **More** total primary production at the base of the food web

- B) Less primary production because carnivores are 100% efficient
- C) The same amount of energy at every level, by definition of food chains
- D) Decomposers to add energy from the sun to dead matter

24. In an ecosystem, **chemical elements** (e.g. carbon, nitrogen in nutrients) are **recycled**, while the energy fixed by **photosynthesis** is eventually:

- A) Re-used indefinitely by the same photon inside every cell with no loss
- B) **Lost** largely as **heat** along the way; not recycled the way a carbon atom can be
- C) Only stored in rocks, not in life
- D) Transferred 100% from producer to all consumers combined

25. Fungi and bacteria in soil return nutrients from dead material to **forms** plants can use again. In contrast to energy, this recycling pattern supports the idea that:

- A) Energy cycles exactly like water in the stratosphere only
 - B) Decomposers are central to **nutrient** cycling for producers, while **energy** flows in one direction through the system
 - C) Producers do not use soil nutrients
 - D) Decomposers are always apex predators in every web
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Part B: Fill in the Blank (10 questions)

Word Bank: Allopatric, allele frequencies, decomposers, density-dependent, gene flow, homologous, K, prezygotic, stabilizing selection, sympatric

26. A population's long-term, stable ceiling in a logistic (S) model is the carrying capacity, symbol _____.

27. The formal focus of **microevolution** is a change in _____ over generations in a population.

28. A barrier that works **before** a zygote—such as different breeding times—is a _____ barrier to reproduction (category name).

29. A limiting factor (such as a contagious disease) whose impact on survival or reproduction can intensify with crowding is often _____.

30. The movement of alleles into or out of a population through migration and mating is called _____.

31. The forelimb bone pattern shared by many tetrapods with different functions (human arm, bat wing, whale flipper) is _____ structure (in comparative anatomy, from common ancestry with modification).

32. Fungi and many bacteria, which break down dead material and return elements to the soil, are _____ in nutrient cycling (category name).

33. Modes of natural selection: when the **middle** of a trait's range is favored and the extremes are selected against, the pattern is often _____.

34. A river splits a continuous population; the two sides evolve in isolation. That geographic mode of speciation is often _____ speciation.

35. A new species arises **without** geographic separation, as in some plant **polyploidy** lineages, is a textbook example of _____ speciation.

Part C: Short Answer (4 questions)

36. (Module 12) In two or three sentences, define **evolutionary fitness** as used in this course, and name **one** line of **evidence** for common descent (fossils, comparative anatomy, biogeography, or molecular) with a brief what-it-shows.

37. (Module 13) Distinguish **bottleneck** and **founder** effects. Then name the three **modes of natural selection** on a continuous trait (directional, stabilizing, disruptive) in one line each.

38. (Module 14) Give one **prezygotic** and one **postzygotic** example of a **reproductive barrier** (one sentence each). In one sentence, contrast **allopatric** and **sympatric** speciation.

39. (Module 15) Contrast **density-dependent** and **density-independent** limiting factors with one **example of each** (one sentence on each type). In two sentences, use the **~10%** energy rule to explain **why** food chains with many links support **fewer** apex consumers than a short chain from the same primary production.

40. (Module 16) Build a short systems explanation: choose one biological scenario from this unit and connect **one evolutionary idea**, **one population/ecology idea**, and **one piece of evidence** into a single cause-and-effect explanation.

End of Practice Test 04